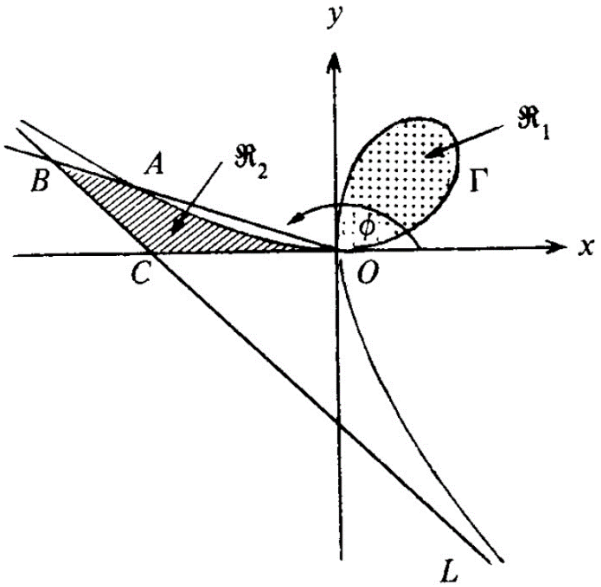
Vector Calculus and Complex Analysis Homework 4

1.



In the figure, O is the origin and Γ is the curve whose equation is $x^3 + y^3 = 3axy$ $(a > 0)$. L is the asymptote of Γ.

(a) Evaluate $\lim_{x \to \infty} \frac{y}{x}$, where $(x, y) \in \Gamma$.

(You may assume that $\lim_{x \to \infty} \frac{y}{x}$ exists.)

Hence, or otherwise, show that the equation of L is

$$x + y + a = 0 .$$

(b) Find the polar equations of Γ and L.

(c) Find the area of the region enclosed by Γ (i.e. $\Re_1$).

(d) Suppose a straight line through O cuts Γ at A and L at B in the second quadrant. Let ϕ be the angle between OB and the positive x-axis. Let A_ϕ be the area of the region bounded by the x-axis, Γ, L and AB (i.e. $\Re_2$).

Show that $A_\phi = \frac{a^2}{2}\left\{\frac{1}{1 + \tan\phi} - \frac{3}{1 + \tan^3\phi} + 2\right\}$.

Solution

1. (a) $x^3 + y^3 = 3axy$

$$\Rightarrow \quad 1 + \left(\frac{y}{x}\right)^3 = 3\left(\frac{a}{x^2}\right)\left(\frac{y}{x}\right)$$

Let $\lim_{x\to\infty} \frac{y}{x} = m$

Taking $x \to \infty$, we have

$$1 - m^3 = 0 \cdot m \quad (\because \lim_{x\to\infty} \frac{a}{x^2} = 0)$$

$$\Rightarrow \quad m = -1$$

Now $x^3 + y^3 = 3axy$

$$\Rightarrow \quad x + y = \frac{3axy}{x^2 - xy + y^2}$$

$$= \frac{3a\left(\frac{y}{x}\right)}{1 - \frac{y}{x} + \left(\frac{y}{x}\right)^2}$$

$$\to \frac{3a(-1)}{1 - (-1) + (-1)^2} = -a \text{ as } x \to \infty$$

$\therefore$ Equation of L is $x + y + a = 0$

(b) Putting $x = r\cos\theta$ and $y = r\sin\theta$,

we have equation of Γ written as

$$r^3\cos^3\theta + r^3\sin^3\theta = 3ar^2\cos\theta\sin\theta$$

i.e. $$r = \frac{3a\cos\theta\sin\theta}{\cos^3\theta + \sin^3\theta}$$

and equation of L written as

$$r\cos\theta + r\sin\theta + a = 0$$

i.e. $$r = \frac{-a}{\cos\theta + \sin\theta}$$

(c) Area of loop $= \frac{1}{2}\int_0^{\frac{\pi}{2}} r^2\,d\theta$

$$= \frac{1}{2}\int_0^{\frac{\pi}{2}} \frac{9a^2\cos^2\theta\sin^2\theta}{(\cos^3\theta+\sin^3\theta)^2}\,d\theta$$

$$= \frac{9}{2}a^2\int_0^{\frac{\pi}{2}} \frac{\tan^2\theta\sec^2\theta}{(1+\tan^3\theta)^2}\,d\theta$$

$$= \frac{9}{2}a^2\int_1^{\infty} \frac{dw}{3w^2} \quad \text{where } w = 1+\tan^3\theta$$

$$= \frac{3}{2}a^2\left[-\frac{1}{w}\right]_1^{\infty}$$

$$= \frac{3}{2}(0+1)a^2$$

$$= \frac{3}{2}a^2$$

(d) $A_\phi = \int_\phi^{\pi} \frac{1}{2}r_1^2 - \frac{1}{2}r_2^2\,d\theta$ where $r_1 = \frac{-a}{\cos\theta+\sin\theta}$

and $r_2 = \frac{3a\cos\theta\sin\theta}{\cos^3\theta+\sin^3\theta}$

$$= \frac{a^2}{2}\left\{\int_\phi^{\pi} \frac{d\theta}{(\sin\theta+\cos\theta)^2} - \int_\phi^{\pi} \frac{9\sin^2\theta\cos^2\theta}{(\sin^3\theta+\cos^3\theta)^2}\,d\theta\right\}$$

Now $\int_\phi^{\pi} \frac{1}{(\sin\theta+\cos\theta)^2}\,d\theta = \int_\phi^{\pi} \frac{\sec^2\theta\,d\theta}{(1+\tan\theta)^2}$

$$= -\left[\frac{1}{1+\tan\theta}\right]_\phi^{\pi}$$

$$= \frac{1}{1+\tan\phi} - 1$$

and $\int_\phi^{\pi} \frac{9\sin^2\theta\cos^2\theta\,d\theta}{(\sin^3\theta+\cos^3\theta)^2} = 9\cdot\left[\frac{1}{-3(1+\tan^3\theta)}\right]_\phi^{\pi}$

$$= 3\left(\frac{1}{1+\tan^3\phi} - 1\right)$$

$$\therefore A_\phi = \frac{a^2}{2}\left\{\left(\frac{1}{1+\tan\phi} - 1\right) - 3\left(\frac{1}{1+\tan^3\phi} - 1\right)\right\}$$

$$\lim_{\phi \to -\frac{3\pi}{4}} A_\phi = \lim_{\phi \to -\frac{3\pi}{4}} \frac{a^2}{2}\left\{\frac{1}{1+\tan\phi} - \frac{3}{1+\tan^3\phi} + 2\right\}$$

$$= a^2 + \frac{a^2}{2}\lim_{\phi \to -\frac{3\pi}{4}}\left\{\frac{1}{1+\tan\phi} - \frac{3}{1+\tan^3\phi}\right\}$$

$$= a^2 + \frac{a^2}{2}\lim_{\phi \to -\frac{3\pi}{4}}\frac{1-\tan\phi+\tan^2\phi-3}{1+\tan^3\phi}$$

$$= a^2 + \frac{a^2}{2}\lim_{\phi \to -\frac{3\pi}{4}}\frac{\tan^2\phi-\tan\phi-2}{1+\tan^3\phi}$$

$$= a^2 + \frac{a^2}{2}\lim_{\phi \to -\frac{3\pi}{4}}\frac{(\tan\phi+1)(\tan\phi-2)}{1+\tan^3\phi}$$

$$= a^2 + \frac{a^2}{2}\lim_{\phi \to -\frac{3\pi}{4}}\frac{\tan\phi-2}{1-\tan\phi+\tan^2\phi}$$

$$= a^2 + \frac{a^2}{2}\left(\frac{(-1)-2}{1-(-1)+(-1)^2}\right)$$

$$= a^2 + \frac{a^2}{2}\left(\frac{-3}{3}\right)$$

$$= \frac{a^2}{2}$$

2. Use polar coordinates to evaluate the double integral.

(a) $\iint_R e^{-\sqrt{x^2+y^2}}\,dA$, where R is the disk $x^2 + y^2 \le 1$.

(b) $\iint_R \cos\sqrt{x^2+y^2}\,dA$, where R is the disk $x^2 + y^2 \le 9$.

(c) $\iint_R (x^2+y^2)dA$, where R is bounded by $y = x$, $y = 0$ and $x = 2$.

(d) $\iint_R x\,dA$, where R is bounded by $r = 1 - \sin\theta$

Solution

(a) $\iint_R e^{-\sqrt{x^2+y^2}}\,dA$, where R is the disk $x^2+y^2 \le 1$.

$$\begin{aligned}
&\iint_R e^{-\sqrt{x^2+y^2}}\,dA \\
&= \int_0^{2\pi}\int_0^1 e^{-r} r\,dr\,d\theta \\
&= \int_0^{2\pi} \left[-(1+r)e^{-r}\right]_{r=0}^{r=1} d\theta \\
&= \int_0^{2\pi} \left(1-2e^{-1}\right) d\theta \\
&= 2\pi - 4\pi e^{-1}
\end{aligned}$$

(b) $\iint_R \cos\sqrt{x^2+y^2}\,dA$, where R is the disk $x^2+y^2 \le 9$.

$$\begin{aligned}
&\iint_R \cos\sqrt{x^2+y^2}\,dA \\
&= \int_0^{2\pi}\int_0^3 r\cos r\,dr\,d\theta \\
&= \int_0^{2\pi} \left[r\sin r + \cos r\right]_{r=0}^{r=3} d\theta \\
&= \int_0^{2\pi} \left(3\sin 3 + \cos 3 - 1\right) d\theta \\
&= 2\pi\left(3\sin 3 + \cos 3 - 1\right)
\end{aligned}$$

(c) $\iint_R (x^2+y^2)dA$, where R is bounded by $y=x$, $y=0$ and $x=2$.

Either polar coordinates or Cartesian coordinates work in this problem (although, the integrals are a bit easier using Cartesian coordinates).

In polar coordinates, $x = 2$ converts to $r\cos\theta = 2$ or $r = 2\sec\theta$, $\theta \in [0, \pi/4]$. We use the table of integrals to evaluate the final integral.

$$\iint_R (x^2+y^2)\,dA = \iint_R r^2\,dA$$
$$= \int_0^{\pi/4}\int_0^{2\sec\theta} r^3\,dr\,d\theta$$
$$= \int_0^{\pi/4} 4\sec^4\theta\,d\theta = 4\int_0^{\pi/4}\sec^4\theta\,d\theta$$
$$= 4\left(\frac{\sec^2\theta\tan\theta}{3}\Big|_0^{\pi/4} + \frac{2}{3}\int_0^{\pi/4}\sec^2\theta\,d\theta\right)$$
$$= 4\left(\left[\frac{\sec^2\theta\tan\theta}{3}\right]_0^{\pi/4} + \left[\frac{2}{3}\tan\theta\right]_0^{\pi/4}\right)$$
$$= 4\left(\frac{2}{3}+\frac{2}{3}\right) = \frac{16}{3}$$

Using Cartesian coordinates.

$$\iint_R (x^2+y^2)\,dA = \int_0^2\int_0^x (x^2+y^2)\,dy\,dx$$
$$= \int_0^2 \frac{4x^3}{3}\,dx = \frac{16}{3}$$

(d) $\iint_R x dA$, where R is bounded by $r = 1 - \sin\theta$

$$\iint_R x \, dA$$
$$= \int_0^{2\pi} \int_0^{1-\sin\theta} r^2 \cos\theta \, dr \, d\theta$$
$$= \frac{1}{3} \int_0^{2\pi} \left[r^3 \cos\theta \right]_{r=0}^{r=1-\sin\theta} d\theta$$
$$= \frac{1}{3} \int_0^{2\pi} (1 - \sin\theta)^3 \cos\theta \, d\theta$$

$$= -\frac{1}{12} \left[(1 - \sin\theta)^4 \right]_0^{2\pi} = 0$$

3. Evaluate the following triple integrals

(a) $\iiint_Q e^{(x^2+y^2+z^2)^{3/2}} dV$, where Q is bounded by the hemisphere $z = \sqrt{4 - x^2 - y^2}$ and the xy-plane.

(b) $\iiint_Q \sqrt{x^2 + y^2 + z^2} dV$, where Q is bounded by $z = \sqrt{x^2 + y^2}$ and $z = \sqrt{2 - x^2 - y^2}$.

(c) $\iiint_Q (x^2 + y^2) dV$, where Q is bounded by the hemisphere $z = 4 - x^2 - y^2$ and the xy-plane.

(d) $\iiint_Q \sqrt{x^2 + y^2 + z^2} dV$, where Q is bounded by the hemisphere $z = -\sqrt{9 - x^2 - y^2}$ and the xy-plane.

(e) $\int_{-2}^{2} \int_{0}^{\sqrt{4-x^2}} \int_{-\sqrt{4-x^2-y^2}}^{0} e^{\sqrt{x^2+y^2+z^2}} dz dy dx$

Solution

(a) $\iiint_Q e^{(x^2+y^2+z^2)^{3/2}}\,dV$, where Q is bounded by the hemisphere $z=\sqrt{4-x^2-y^2}$ and the xy-plane.

$$\iiint_Q e^{(x^2+y^2+z^2)^{3/2}}\,dV$$

$$= \int_0^{2\pi}\int_0^{\pi/2}\int_0^2 e^{\rho^3}\rho^2\sin\phi\;d\rho\,d\phi\,d\theta$$

$$= 2\pi\int_0^{\pi/2}\int_0^2 e^{\rho^3}\rho^2\sin\phi\;d\rho\,d\phi$$

$$= 2\pi\int_0^{\pi/2}\frac{1}{3}(e^8-1)\sin\phi\;d\rho\,d\phi$$

$$= (2\pi)\frac{1}{3}(e^8-1) = \frac{2\pi}{3}(e^8-1)$$

(b) $\iiint_Q \sqrt{x^2+y^2+z^2}\,dV$, where Q is bounded by $z=\sqrt{x^2+y^2}$ and $z=\sqrt{2-x^2-y^2}$.

The solid is the portion of the ball with $0 \le \phi \le \pi/4$.

$$\iiint_Q \sqrt{x^2+y^2+z^2}\,dV$$
$$= \int_0^{2\pi}\int_0^{\pi/4}\int_0^{\sqrt{2}} \rho^3 \sin\phi\, d\rho\, d\phi\, d\theta$$
$$= 2\pi \int_0^{\pi/4}\int_0^{\sqrt{2}} \rho^3 \sin\phi\, d\rho\, d\phi$$
$$= 2\pi \int_0^{\pi/4} \sin\phi\, d\phi$$
$$= 2\pi\left(1-\frac{1}{\sqrt{2}}\right) = (2-\sqrt{2})\pi$$

(c) $\iiint_Q (x^2+y^2)dV$, where Q is bounded by the hemisphere $z=4-x^2-y^2$ and the xy-plane.

Here we use cylindrical coordinates.

$$\iiint_Q (x^2+y^2)\,dV$$
$$= \int_0^{2\pi}\int_0^{2}\int_0^{4-r^2} r^3\,dz\,dr\,d\theta$$
$$= 2\pi \int_0^{2}\int_0^{4-r^2} r^3\,dz\,dr$$
$$= 2\pi \int_0^{2} (4-r^2)r^3\,dr = \frac{32\pi}{3}$$

(d) $\iiint_Q \sqrt{x^2+y^2+z^2}\,dV$, where Q is bounded by the hemisphere $z=-\sqrt{9-x^2-y^2}$ and the xy-plane.

$$\iiint_Q \sqrt{x^2+y^2+z^2}\,dV$$
$$= \int_0^{2\pi}\int_{\pi/2}^{\pi}\int_0^3 \rho^3 \sin\phi\; d\rho\, d\phi\, d\theta$$
$$= \int_0^{2\pi}\int_{\pi/2}^{\pi} \frac{81}{4}\sin\phi\; d\phi\, d\theta$$
$$= \int_0^{2\pi} \frac{81}{4}\, d\theta = \frac{81\pi}{2}$$

(e) $\int_{-2}^{2}\int_0^{\sqrt{4-x^2}}\int_{-\sqrt{4-x^2-y^2}}^{0} e^{\sqrt{x^2+y^2+z^2}}\,dz dy dx$

$$I = \int_{-2}^{2}\int_0^{\sqrt{4-x^2}}\int_{-\sqrt{4-x^2-y^2}}^{0} e^{\sqrt{x^2+y^2+z^2}}\,dz\,dy\,dx$$

$$= \int_0^{\pi}\int_{\frac{\pi}{2}}^{\pi}\int_0^2 e^{\rho}\rho^2\sin\phi\; d\rho d\phi\, d\theta$$
$$= \int_0^{\pi}\int_{\frac{\pi}{2}}^{\pi} \left[\rho^2 e^{\rho} - 2\rho e^{\rho} + 2e^{\rho}\right]_0^2 \sin\phi d\phi\, d\theta$$
$$= \int_0^{\pi}\int_{\frac{\pi}{2}}^{\pi} 2\left(e^2-1\right)\sin\phi\, d\phi\, d\theta$$
$$= 2\left(e^2-1\right)\int_0^{\pi}\left[-\cos\phi\right]_{\frac{\pi}{2}}^{\pi} d\theta$$
$$= 2\left(e^2-1\right)\int_0^{\pi} d\theta$$
$$= 2\left(e^2-1\right)\pi$$